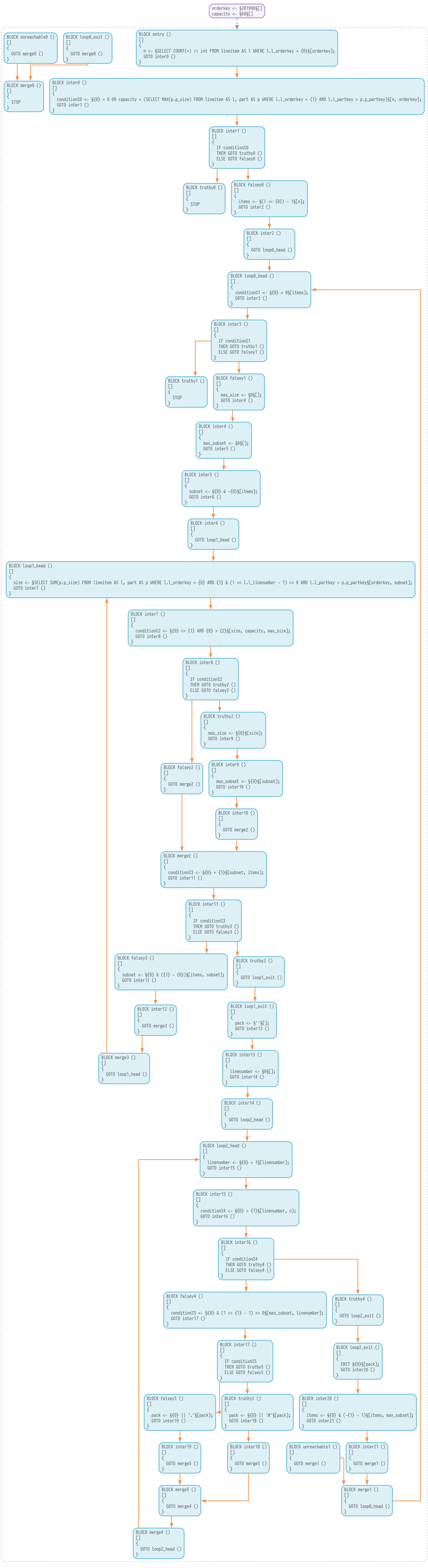
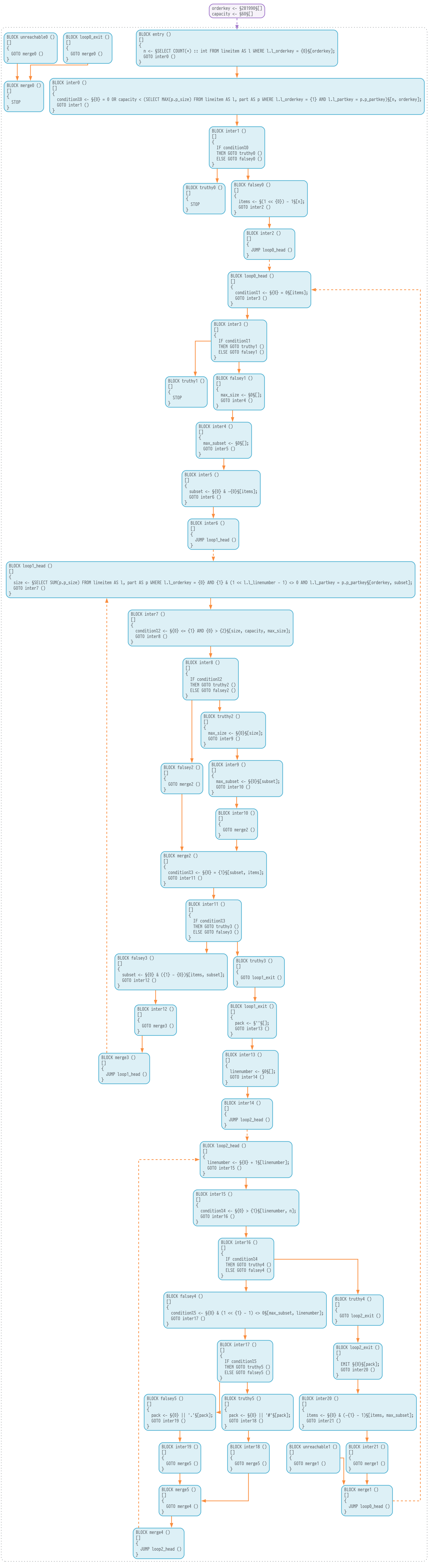
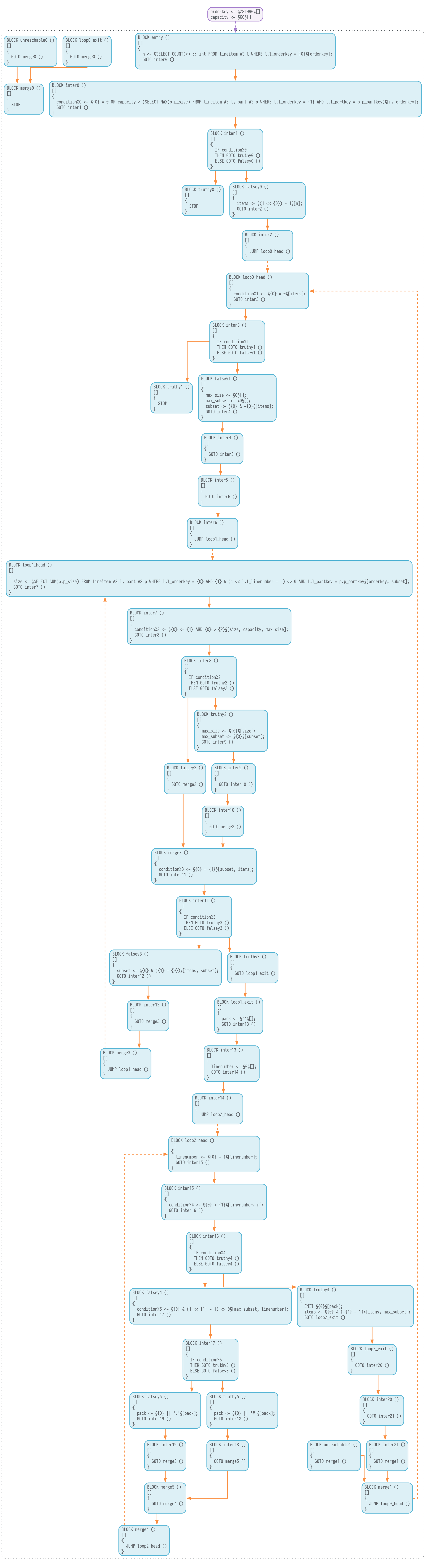
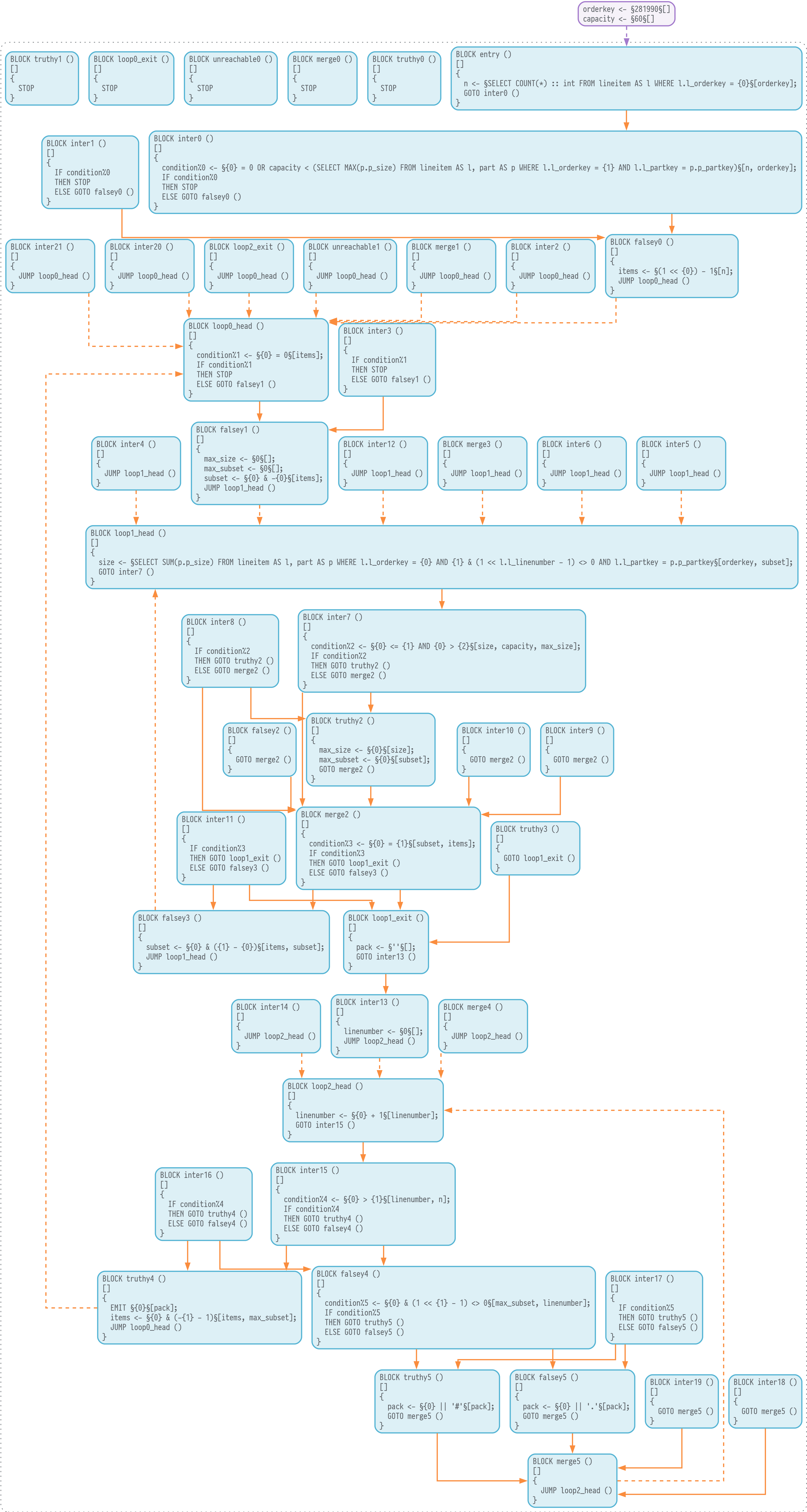
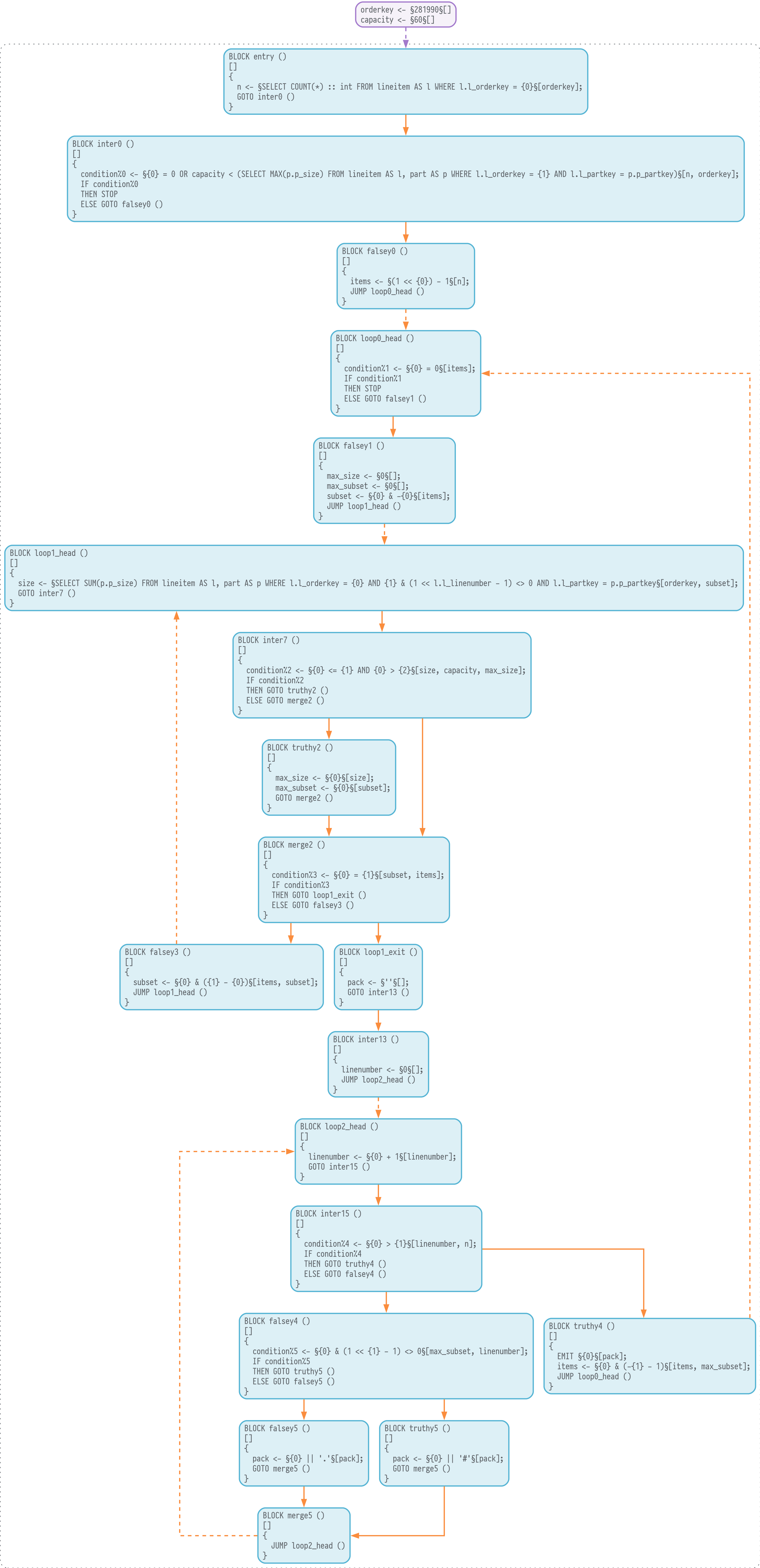


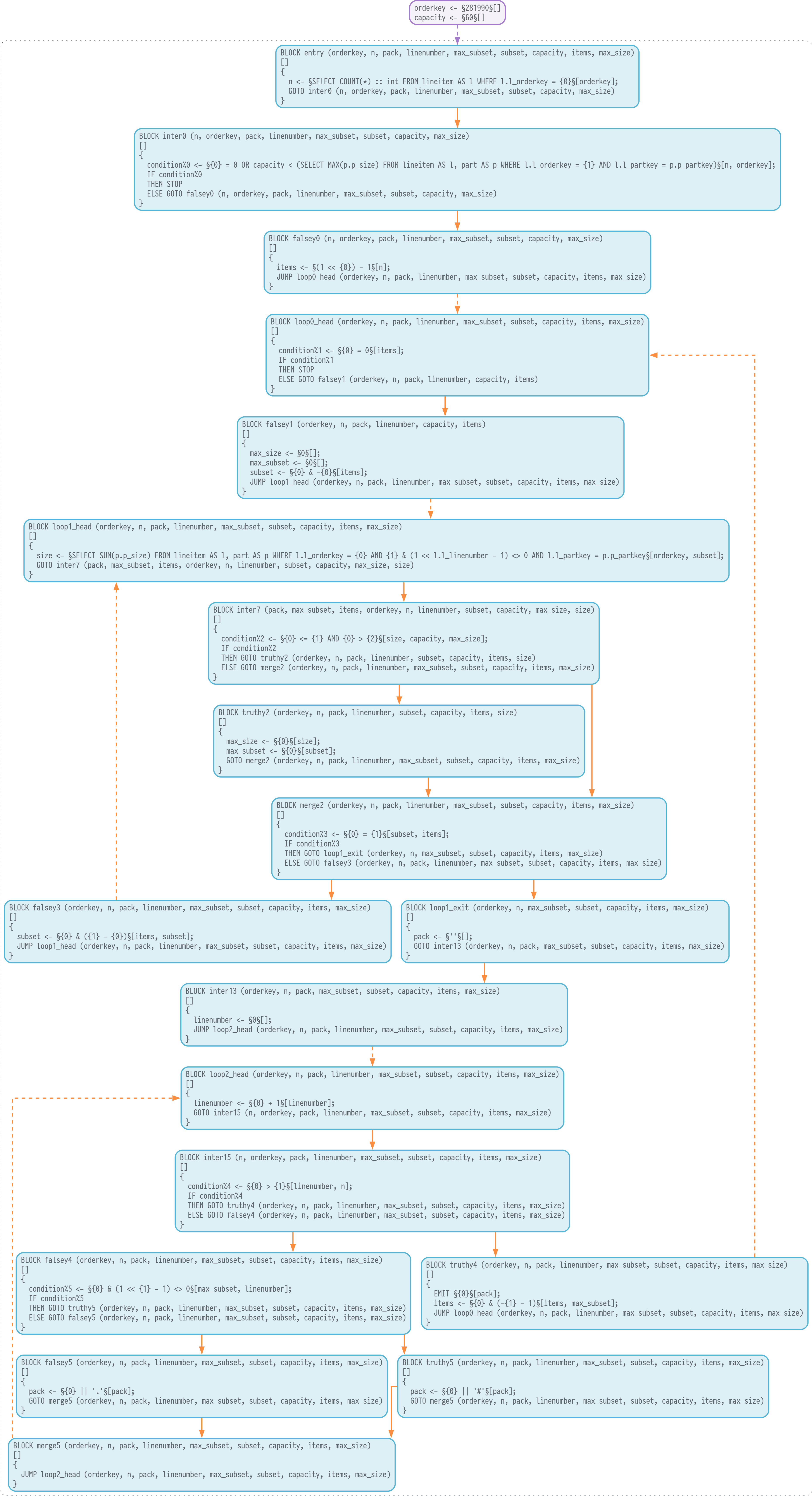













```
orderkey <- $281990$[]
capacity <- $60$[]
n <- $NULL$[]
pack <- $NULL$[]
linenumber <- $NULL$[]
max_subset <- $NULL$[]
subset <- $NULL$[]
items <- $NULL$[]
max_size <- $NULL$[]
```

```
BLOCK entry (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[]
{
  n <- $SELECT COUNT(*) :: int FROM lineitem AS l WHERE l.l_orderkey = {0}$[orderkey];
  GOTO inter0 (n, orderkey, pack, linenumber, max_subset, subset, capacity, max_size)
}
```

```
BLOCK inter0 (n, orderkey, pack, linenumber, max_subset, subset, capacity, max_size)
[]
{
  condition%0 <- ${0} = 0 OR capacity < (SELECT MAX(p.p_size) FROM lineitem AS l, part AS p WHERE l.l_orderkey = {1} AND l.l_partkey = p.p_partkey)$[n, orderkey];
  IF condition%0
  THEN STOP
  ELSE GOTO falsey0 (n, orderkey, pack, linenumber, max_subset, subset, capacity, max_size)
}
```

```
BLOCK falsey0 (n, orderkey, pack, linenumber, max_subset, subset, capacity, max_size)
[]
{
  items <- ${1 << {0}} - 1$[n];
  JUMP loop0_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK loop0_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[]
{
  condition%1 <- ${0} = 0$[items];
  IF condition%1
  THEN STOP
  ELSE GOTO falsey1 (orderkey, n, pack, linenumber, capacity, items)
}
```

```
BLOCK falsey1 (orderkey, n, pack, linenumber, capacity, items)
[]
{
  max_size <- $0$[];
  max_subset <- $0$[];
  subset <- ${0} & ~{0}$[items];
  JUMP loop1_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK loop1_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[]
{
  size <- $SELECT SUM(p.p_size) FROM lineitem AS l, part AS p WHERE l.l_orderkey = {0} AND {1} & (1 << l.l_linenumber - 1) <> 0 AND l.l_partkey = p.p_partkey$[orderkey, subset];
  GOTO inter7 (pack, max_subset, items, orderkey, n, linenumber, subset, capacity, max_size, size)
}
```

```
BLOCK inter7 (pack, max_subset, items, orderkey, n, linenumber, subset, capacity, max_size, size)
[]
{
  condition%2 <- ${0} <= {1} AND {0} > {2}$[size, capacity, max_size];
  IF condition%2
  THEN GOTO truthy2 (orderkey, n, pack, linenumber, subset, capacity, items, size)
  ELSE GOTO merge2 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK truthy2 (orderkey, n, pack, linenumber, subset, capacity, items, size)
[]
{
  max_size <- ${0}$[size];
  max_subset <- ${0}$[subset];
  GOTO merge2 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK merge2 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[]
{
  condition%3 <- ${0} = {1}$[subset, items];
  IF condition%3
  THEN GOTO loop1_exit (orderkey, n, max_subset, subset, capacity, items, max_size)
  ELSE GOTO falsey3 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK falsey3 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[]
{
  subset <- ${0} & ({1} - {0})$[items, subset];
  JUMP loop1_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK loop1_exit (orderkey, n, max_subset, subset, capacity, items, max_size)
[]
{
  pack <- '$'[pack];
  GOTO inter13 (orderkey, n, pack, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK inter13 (orderkey, n, pack, max_subset, subset, capacity, items, max_size)
[]
{
  linenumber <- $0$[];
  JUMP loop2_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK loop2_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[]
{
  linenumber <- ${0} + 1$[linenumber];
  GOTO inter15 (n, orderkey, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK inter15 (n, orderkey, pack, linenumber, max_subset, subset, capacity, items, max_size)
[]
{
  condition%4 <- ${0} > {1}$[linenumber, n];
  IF condition%4
  THEN GOTO truthy4 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
  ELSE GOTO falsey4 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK falsey4 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[]
{
  condition%5 <- ${0} & (1 << {1} - 1) <> 0$[max_subset, linenumber];
  IF condition%5
  THEN GOTO truthy5 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
  ELSE GOTO falsey5 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK truthy4 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[]
{
  EMIT ${0}$[pack];
  items <- ${0} & (~{1} - 1)$[items, max_subset];
  JUMP loop0_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK falsey5 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[]
{
  pack <- ${0} || '.'$[pack];
  GOTO merge5 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK truthy5 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[]
{
  pack <- ${0} || '#'$[pack];
  GOTO merge5 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK merge5 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[]
{
  JUMP loop2_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```



```
orderkey <- $281990$[]
capacity <- $60$[]
n <- $NULL$[]
pack <- $NULL$[]
linenumber <- $NULL$[]
max_subset <- $NULL$[]
subset <- $NULL$[]
items <- $NULL$[]
max_size <- $NULL$[]
```

```
BLOCK entry (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[
  FROM LOOP AS entry
]
{
  n <- $SELECT COUNT(*) :: int FROM lineitem AS l WHERE l.l_orderkey = {0}$[orderkey];
  GOTO inter0 (n, orderkey, pack, linenumber, max_subset, subset, capacity, max_size)
}
```

```
BLOCK inter0 (n, orderkey, pack, linenumber, max_subset, subset, capacity, max_size)
[
  FROM entry
  WITH (n, orderkey, pack, linenumber, max_subset, subset, capacity, max_size)
  WHERE TRUE
]
{
  condition%0 <- ${0} = 0 OR capacity < (SELECT MAX(p.p_size) FROM lineitem AS l, part AS p WHERE l.l_orderkey = {1} AND l.l_partkey = p.p_partkey)$[n, orderkey];
  IF condition%0
  THEN STOP
  ELSE GOTO falsey0 (n, orderkey, pack, linenumber, max_subset, subset, capacity, max_size)
}
```

```
BLOCK falsey0 (n, orderkey, pack, linenumber, max_subset, subset, capacity, max_size)
[
  FROM inter0
  WITH (n, orderkey, pack, linenumber, max_subset, subset, capacity, max_size)
  WHERE (TRUE) AND (NOT (condition%0))
]
{
  items <- ${1 << {0}} - 1$[n];
  JUMP loop0_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK loop0_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[
  FROM LOOP AS loop0_head
]
{
  condition%1 <- ${0} = 0$[items];
  IF condition%1
  THEN STOP
  ELSE GOTO falsey1 (orderkey, n, pack, linenumber, capacity, items)
}
```

```
BLOCK falsey1 (orderkey, n, pack, linenumber, capacity, items)
[
  FROM loop0_head
  WITH (orderkey, n, pack, linenumber, capacity, items)
  WHERE (TRUE) AND (NOT (condition%1))
]
{
  max_size <- $0$[];
  max_subset <- $0$[];
  subset <- ${0} & --{0}$[items];
  JUMP loop1_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK loop1_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[
  FROM LOOP AS loop1_head
]
{
  size <- $SELECT SUM(p.p_size) FROM lineitem AS l, part AS p WHERE l.l_orderkey = {0} AND {1} & (1 << l.l_linenumber - 1) <> 0 AND l.l_partkey = p.p_partkey$[orderkey, subset];
  GOTO inter7 (pack, max_subset, items, orderkey, n, linenumber, subset, capacity, max_size, size)
}
```

```
BLOCK inter7 (pack, max_subset, items, orderkey, n, linenumber, subset, capacity, max_size, size)
[
  FROM loop1_head
  WITH (pack, max_subset, items, orderkey, n, linenumber, subset, capacity, max_size, size)
  WHERE TRUE
]
{
  condition%2 <- ${0} <= {1} AND {0} > {2}$[size, capacity, max_size];
  IF condition%2
  THEN GOTO truthy2 (orderkey, n, pack, linenumber, subset, capacity, items, size)
  ELSE GOTO merge2 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK truthy2 (orderkey, n, pack, linenumber, subset, capacity, items, size)
[
  FROM inter7
  WITH (orderkey, n, pack, linenumber, subset, capacity, items, size)
  WHERE (TRUE) AND (condition%2)
]
{
  max_size <- ${0}$[size];
  max_subset <- ${0}$[subset];
  GOTO merge2 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK merge2 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[
  FROM inter7
  WITH (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
  WHERE (TRUE) AND (NOT (condition%2)),
  FROM truthy2
  WITH (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
  WHERE TRUE
]
{
  condition%3 <- ${0} = {1}$[subset, items];
  IF condition%3
  THEN GOTO loop1_exit (orderkey, n, max_subset, subset, capacity, items, max_size)
  ELSE GOTO falsey3 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK falsey3 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[
  FROM merge2
  WITH (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
  WHERE (TRUE) AND (NOT (condition%3))
]
{
  subset <- ${0} & ({1} - {0})$[items, subset];
  JUMP loop1_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK loop1_exit (orderkey, n, max_subset, subset, capacity, items, max_size)
[
  FROM merge2
  WITH (orderkey, n, max_subset, subset, capacity, items, max_size)
  WHERE (TRUE) AND (condition%3)
]
{
  pack <- '$'[];
  GOTO inter13 (orderkey, n, pack, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK inter13 (orderkey, n, pack, max_subset, subset, capacity, items, max_size)
[
  FROM loop1_exit
  WITH (orderkey, n, pack, max_subset, subset, capacity, items, max_size)
  WHERE TRUE
]
{
  linenumber <- $0$[];
  JUMP loop2_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK loop2_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[
  FROM LOOP AS loop2_head
]
{
  linenumber <- ${0} + 1$[linenumber];
  GOTO inter15 (n, orderkey, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK inter15 (n, orderkey, pack, linenumber, max_subset, subset, capacity, items, max_size)
[
  FROM loop2_head
  WITH (n, orderkey, pack, linenumber, max_subset, subset, capacity, items, max_size)
  WHERE TRUE
]
{
  condition%4 <- ${0} > {1}$[linenumber, n];
  IF condition%4
  THEN GOTO truthy4 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
  ELSE GOTO falsey4 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK falsey4 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[
  FROM inter15
  WITH (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
  WHERE (TRUE) AND (NOT (condition%4))
]
{
  condition%5 <- ${0} & (1 << {1} - 1) <> 0$[max_subset, linenumber];
  IF condition%5
  THEN GOTO truthy5 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
  ELSE GOTO falsey5 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK truthy4 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[
  FROM inter15
  WITH (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
  WHERE (TRUE) AND (condition%4)
]
{
  EMIT ${0}$[pack];
  items <- ${0} & (--{1} - 1)$[items, max_subset];
  JUMP loop0_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK falsey5 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[
  FROM falsey4
  WITH (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
  WHERE (TRUE) AND (NOT (condition%5))
]
{
  pack <- ${0} || '.'$[pack];
  GOTO merge5 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK truthy5 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[
  FROM falsey4
  WITH (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
  WHERE (TRUE) AND (condition%5)
]
{
  pack <- ${0} || '#'$[pack];
  GOTO merge5 (orderkey, n, pack, max_subset, subset, capacity, items, max_size)
}
```

```
BLOCK merge5 (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
[
  FROM truthy5
  WITH (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
  WHERE TRUE,
  FROM falsey5
  WITH (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
  WHERE TRUE
]
{
  JUMP loop2_head (orderkey, n, pack, linenumber, max_subset, subset, capacity, items, max_size)
}
```